

AhamX Bodhi

Where I Becomes Infinite

CLIP and Contrastive Learning

Module 4.1: Vision-Language Models

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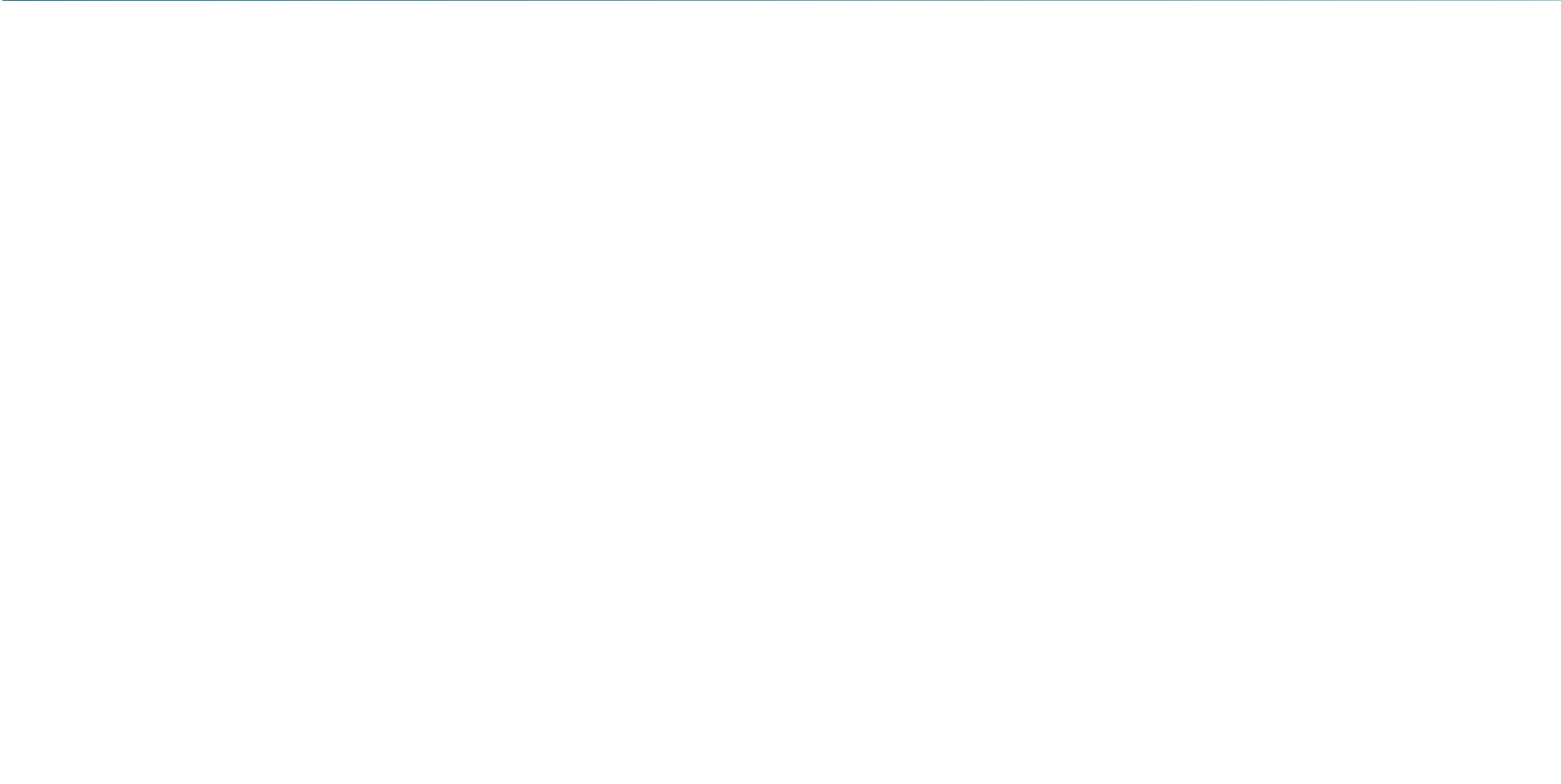
Session Overview

What You Will Learn

- What contrastive learning is and why it works
- CLIP architecture and training objective
- InfoNCE loss: the math and intuition
- CLIP variants: SigLIP, ALIGN, OpenCLIP
- Production uses for AI engineers

Concept at a Glance

- **Duration:** 10 minutes
- **Bloom's Level:** B3 – Apply
- **Difficulty:** L3 – Advanced
- **Module:** 4.1 Vision-Language Models
- **Key paper:** Radford et al., 2021 (OpenAI)





Contrastive Learning: The Core Intuition

Learning What Things Are By Learning What They Are Not

- **Core idea:** Train a model by comparing *positive pairs* (similar items) against *negative pairs* (dissimilar items) – no explicit class labels needed
- **Objective:** Pull the representations of positive pairs closer together in embedding space; push negative pairs apart
- **Self-supervised power:** Positive pairs can be constructed automatically (augmented views of same image, or matching image-caption pairs) – no human labeling required
- **Learned metric:** The encoder learns a metric space where proximity = semantic similarity – transferable to any downstream task
- **Scale advantage:** With internet-scale data, negatives are abundant and diverse – the model sees every possible way things can differ

Contrastive learning is the training recipe that enables CLIP, DINOv2, SimCLR, and MoCo – all of which learn powerful features without labeled datasets.



CLIP Architecture: Dual-Encoder Design

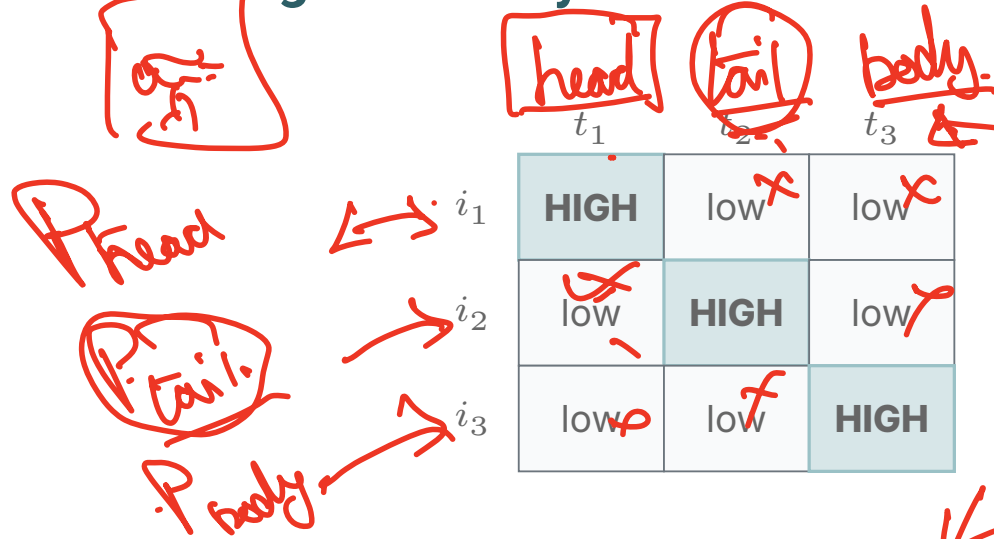
Contrastive Language-Image Pre-Training (Radford et al., 2021)

- **Image encoder:** ViT-B/32, ViT-B/16, ViT-L/14 – produces a single D -dim vector per image via CLS token
- **Text encoder:** 12-layer Transformer with BPE tokenizer – processes caption up to 77 tokens, output is embedding of the [EOS] token
- **Projection heads:** Separate linear projections for image and text embeddings into a shared D -dim space (typically 512 or 768)
- **Training data:** 400 million (image, text) pairs scraped from the internet – WIT (WebImageText) dataset, not publicly released
- **Objective:** For a batch of N pairs, maximize similarity of N matching pairs and minimize similarity of $N^2 - N$ non-matching pairs simultaneously

The dual-encoder design means image and text are encoded *independently* – enabling pre-computation of all image embeddings for fast retrieval at inference time.



CLIP Training: Similarity Matrix



Diagonal = positive pairs (maximize)
Off-diagonal = negative pairs (minimize)

InfoNCE loss: $\mathcal{L} = -\frac{1}{N} \sum_{i=1}^N \log \frac{\exp(\text{sim}(i_i, t_i)/\tau)}{\sum_{j=1}^N \exp(\text{sim}(i_i, t_j)/\tau)}$. Temperature τ controls sharpness.

$$Q = G^{-1} \left(\frac{i_i \cdot b_i}{1 \cdot 1 \quad 1 \cdot 1} \right) = 0$$

$$= 1 \quad \epsilon$$



InfoNCE Loss: The Engineering Perspective

Why Batch Size Matters Critically for CLIP

- **Loss formulation:** InfoNCE (Noise Contrastive Estimation) is a cross-entropy loss over a batch of N pairs – treating negatives as $N - 1$ “distractor” classes
- **Batch size effect:** Larger batches provide more negatives per positive – critical for learning fine-grained distinctions. CLIP used batch size 32,768 across 592 GPUs.
- **Temperature τ :** Learnable temperature parameter (initialized to 0.07). Low τ = sharp distribution (confident), high τ = soft distribution (uncertain).
- **Hard negatives:** In-batch negatives are mostly easy (random image-caption pairs are clearly mismatched). Mining hard negatives (similar but incorrect pairs) significantly improves performance.
- **Memory efficiency:** Gradient checkpointing + mixed precision essential; full $N \times N$ similarity matrix at $N = 32768$ requires careful memory management

The impracticality of large-batch contrastive training on single GPUs motivated SigLIP’s sigmoid loss – which decouples examples and eliminates the batch-size bottleneck.





SigLIP: Sigmoid Loss for Language-Image Pre-Training

Zhai et al., 2023 – Google

- **Problem with InfoNCE:** Requires global normalization over all N pairs in the batch – forces large batches for good performance, limits distributed training efficiency
- **SigLIP solution:** Replace softmax normalization with sigmoid – treat each (image, text) pair as an independent binary classification (match / no match)
- **Loss:** $\mathcal{L} = -\frac{1}{N^2} \sum_{i,j} \log \sigma(z_{ij} \cdot y_{ij} - b)$ where $y_{ij} = +1$ for matching pairs, -1 for non-matching, b is a bias term
- **Result:** Better performance at smaller batch sizes; scales better across distributed hardware; used by Google in InternVL2 and LLaMA-3.2-Vision

Current Standard

SigLIP-SO/400M has replaced CLIP ViT-L/14 as the preferred VLM visual encoder in most new open-source VLMs (LLaVA-OneVision, InternVL2, PaliGemma).

CLIP Ecosystem: Models and Variants



Model	Data	Loss	IN Zero-Shot	Notes
CLIP (OpenAI)	400M	InfoNCE	75.3%	Original, ViT-L/14
OpenCLIP	LAION-2B	InfoNCE	79.2%	Open-source, EVA-CLIP
ALIGN (Google)	1.8B	InfoNCE	76.4%	Noisy data, EfficientNet
SigLIP-SO/400M	WebLI	Sigmoid	83.1%	Current VLM standard
EVA-CLIP-18B	Merged	InfoNCE	83.4%	Largest CLIP, ViT-G
MetaCLIP	CommonPool	InfoNCE	80.5%	Data curation recipe





GenAI Application: Zero-Shot Image Classification

CLIP Zero-Shot: No Training Data Required

- **Mechanism:** Encode image, encode candidate class names as text prompts ("a photo of a {class}"), find the text embedding nearest to the image embedding via cosine similarity
- **Prompt engineering matters:** "a photo of a dog" outperforms "dog" by 3–5% on ImageNet. Ensemble multiple prompts: "a photo of a {c}", "a blurry photo of a {c}", etc.
- **Production use:** Content moderation – classify uploaded images into "safe / unsafe / adult / violence" categories without per-category training
- **OpenAI CLIP example:** Load openai/clip-vit-large-patch14; encode image; encode 1000 class prompts; matrix multiply + softmax = classification probabilities
- **Limitation:** Degraded performance on fine-grained categories, satellite imagery, and domain-specific images not well represented in WIT training data

For best results, always compare zero-shot CLIP with a fine-tuned linear probe before deploying. Linear probe adds only 1 hour of training and often improves accuracy by 10–15%.



GenAI Application: Cross-Modal Search Systems

Text-to-Image and Image-to-Text Search

- **Architecture:** Encode entire image catalog offline with CLIP image encoder; store in vector database (Pinecone, Weaviate, Qdrant); at query time encode text query and retrieve nearest image embeddings
- **Bidirectional:** Same architecture works for image-to-text – encode query image, retrieve nearest text documents. Powers reverse image search.
- **Hybrid search:** Combine CLIP similarity score with BM25 text score for higher precision. Weaviate and Qdrant support hybrid vector+keyword search natively.
- **Scale:** Unsplash uses CLIP to search 3M+ photos by natural language description. Shopify uses CLIP for catalog search across 100M+ product images.

Latency Optimization

- Pre-compute and cache all image embeddings – only text query needs real-time encoding (<5ms)
- HNSW index (ef=200, M=32) retrieves from 10M vectors in <20ms



GenAI Application: CLIP in Image Generation

CLIP as the Semantic Bridge for Diffusion Models

- **Stable Diffusion:** CLIP text encoder (ViT-L/14) converts the text prompt into a sequence of 77 token embeddings that condition the UNet denoiser via cross-attention
- **SDXL and SD3:** Use multiple CLIP text encoders (OpenCLIP ViT-G + CLIP ViT-L) concatenated for richer text conditioning; SD3 adds T5-XXL for long prompts
- **DALL-E 2 prior:** CLIP image embeddings can be used to generate images “in the style of” an input image – the prior maps image embeddings to diffusion conditioning space
- **FLUX.1:** Uses CLIP-L for local features + T5-XXL for semantic understanding – two-encoder approach separates style from content conditioning
- **ControlNet:** CLIP image encoder enables IP-Adapter – inject visual style/content from a reference image into any SD generation

CLIP is the “glue” between language and vision – almost every production text-to-image system relies on CLIP embeddings for text conditioning.



GenAI Application: CLIP Embeddings in Multimodal RAG

Unified Semantic Index with CLIP

- **Idea:** Since CLIP image and text embeddings share the same space, images and text can be stored in the *same vector index* and retrieved with text queries
- **Pipeline:** Index text chunks (via text encoder) and images (via image encoder) together in Weaviate or Qdrant; query with natural language; retrieve mixed text+image results; pass to VLM for answer generation
- **Use case:** Engineering handbook search – user asks “how to replace the hydraulic seal?” and retrieves both the relevant text procedure AND the corresponding diagram
- **Advantage:** No separate image captioning step required – CLIP directly enables text-to-image retrieval in a unified index

Limitation to Know

CLIP's 77-token text context window limits indexing of long text chunks. Use a dedicated text embedding model (e.g., text-embedding-3-large) for text retrieval and CLIP only for image retrieval.



Key Takeaways

What You Should Remember

- Contrastive learning trains encoders by maximizing similarity of matching pairs and minimizing similarity of non-matching pairs – no labels needed at internet scale
- CLIP uses dual-encoder (ViT + Transformer) with InfoNCE loss on 400M image-text pairs to create a shared vision-language embedding space
- SigLIP replaces softmax with sigmoid loss – decouples training examples, enables smaller batch sizes, and is now the standard VLM visual encoder
- CLIP enables zero-shot classification, cross-modal retrieval, and text-based conditioning for diffusion models
- For production: use OpenCLIP ViT-L or SigLIP-SO for VLM visual encoders; CLIP for retrieval; EVA-CLIP-18B for maximum zero-shot accuracy

$[H]$
 $m \times d_{img}$
 $\uparrow$
 $\# \text{ Patches}$

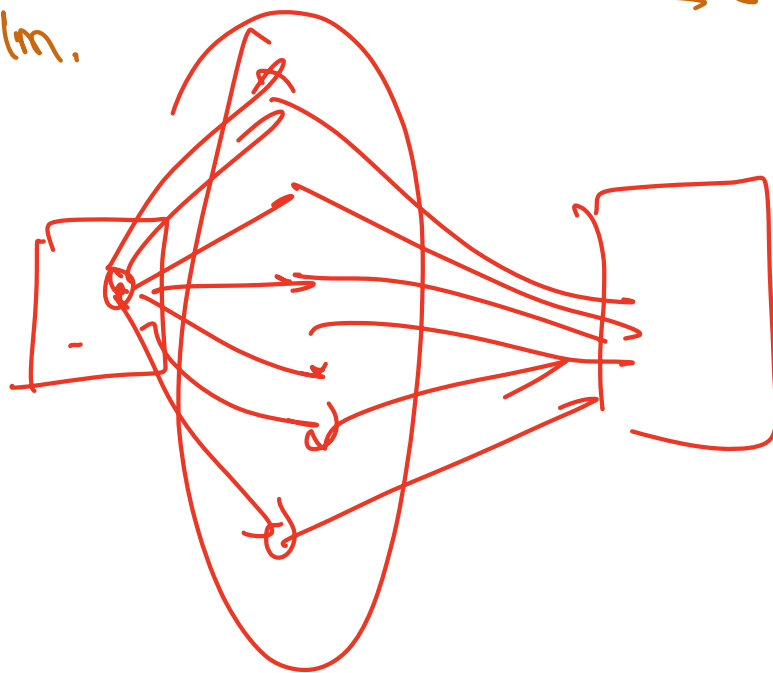
$\longrightarrow$
 $[]$
 $m \times d_{img}$

$\nwarrow$ Connectors

$T_1 \dots T_{k+m}$
 $[C]$
 $k+m$

$[]$
 $m \times d_{img}$
 $m \times d_{img}$

$[P]$
 $d_{img} \times d_{img}$
 $\uparrow$



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